PROJECT METHODOLOGY

Electric Vehicle Charge and Range Analysis

The methodology followed in this project consists of systematic steps from data acquisition to insight generation.

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1. Data Collection

The dataset was collected from publicly available EV data sources. The dataset includes:

- EV brands and models
- Charging station regions and types
- Performance metrics (top speed)
- Efficiency (Wh/Km)
- Pricing details
- Powertrain types
- Body styles

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2. Data Cleaning

Data preprocessing was performed using Microsoft Excel:

- Removed null and duplicate values
- Standardized brand names
- Converted numerical fields to consistent units
- Structured columns properly
- Categorized vehicles by body style

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3. Data Preparation & Transformation

- Calculated average top speed per brand
- Computed average efficiency
- Segmented price categories
- Grouped charging stations by region
- Classified powertrain types

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4. Dashboard Development

The cleaned dataset was imported into Tableau.

The following visualizations were created:

- Charging Infrastructure Map
- Brand & Model Distribution Charts
- Performance Comparison Graphs
- Efficiency Analysis
- Price Segmentation Dashboard
- Powertrain Distribution Charts

Interactive filters were added for dynamic exploration.

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5. Insight Generation

The dashboard was analyzed to extract meaningful insights:

- Identification of high charging concentration regions
- Performance comparison across brands
- Affordability segmentation analysis
- Infrastructure readiness evaluation
- Efficiency variation study

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6. Validation

All visualizations were cross-verified with raw dataset values to ensure accuracy and consistency.

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7. Final Documentation

The entire system workflow, results, findings, and future enhancements were documented systematically.